Homework 4: Binary (15 points)

Due to Gradescope by 11:45 PM on Tuesday, September 20th
You need to submit a pdf to Gradescope; failure to assign questions to pages will result in a 10% deduction on your grade

Homework Goal: Practice binary conversions and operations;
learn more about ancient binary through reading and writing

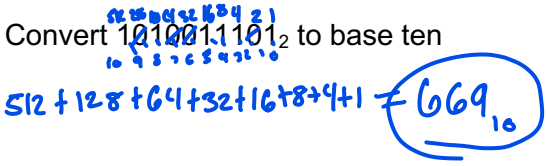
Vocab

1. Practice your vocab from this chapter! (3 points)
   (a) What does "bit" stand for? binary digit (b)
   (b) Overflow occurs when a computer tries to represent a number that exceeds the maximum value.
   (c) Instead of using a decimal point, we use a binary point for fractional numbers in binary.
   (d) To represent text in binary, the computer assigns each printable letter or symbol a unique number called a code mapping.
   (e) The number of bits used to store each sample is referred to as bit depth.
   (f) Sampling occurs at fixed time intervals and is when the signal amplitude is measured/stored, while digitizing is the process of converting an analog signal to a digital number.

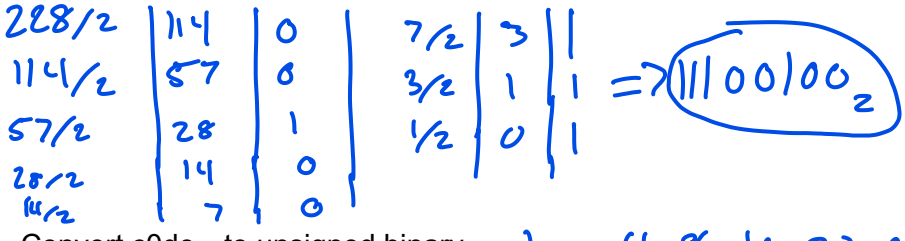
Conversions

Perform the following conversions. *You must show your work for full credit.* (1 point each)

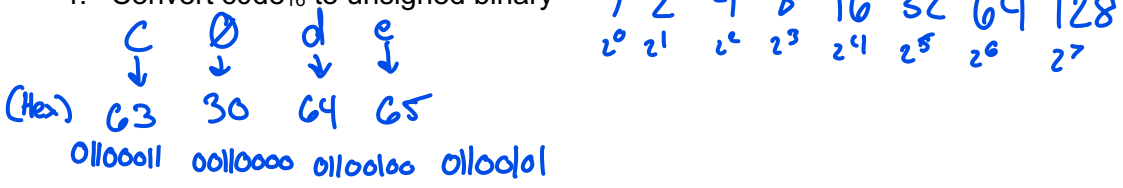
2. Convert 1010011101_2 to base ten



3. Convert 228_{10} to unsigned binary



4. Convert $c0de_{16}$ to unsigned binary



5. Convert the unsigned binary number 1100001011_2 (unsigned) to base 16 (hex)

6 1 11 → B

6 | B

6. Convert 201_7 to decimal

$1 \cdot 7^0 + 0 \cdot 7^1 + 2 \cdot 7^2$

$1 + 0 + 98$

$= 99_{10}$

7. Convert 264_8 to base 4 (easiest to go through base 10)

$2 \cdot 8^2 + 6 \cdot 8^1 + 4 \cdot 8^0$

$= 180_{10}$

$\Rightarrow$

180/4	45	0
45/4	11	1
11/4	2	3
2/4	0	2

$\Rightarrow 2310_4$

8. Convert -30_{10} to *two's complement* binary

11110 $\Rightarrow$ 30

000001

(NEG) ↓ 1 + 1

00010 → 100010

$-32 + 2 = -30$

9. Convert 2.375_{10} to unsigned binary

$2.375 \rightarrow 2 + 2^{-2} + 2^{-3}$

10.011×2^0

1.0011×2^1

$M = 1.011$

$B = 2$

$E = 1$

Operations

10. Perform the following binary arithmetic operations, assuming that we are using *two's complement representation* and only have *6 bits available* for representation. Leave your answer in binary and **state whether or not overflow occurs**. (1 point)

(a) $101011_2 + 000111_2$

101011

+ 000111

110010

Overflow?

NO

(b) $011101_2 + 001110_2$

8/4

```
  1 1 1 1
  0 1 1 1 0 1
+ 0 0 1 1 1 0
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  1 0 1 0 1 1
```

Overflow?

Yes

Data Representation - Numbers

11. Fill in the table with the range of values (give numerical answers for minimum and maximum) that can be represented if *7 bits* are available for these representations: (0.5 points)

notation	minimum	maximum
unsigned binary	0	127
two's complement	-64	63

Data Representation - ASCII

12. Using the ASCII code set at https://www.ascii-code.com (0.5 points)
    (a) Show the internal binary representation of the following **four-character** string: CSCI

C => 01000011 I => 01001001

S => 01010011

    (b) What character does 00100001 represent?

!

Data Representation - Sound & Color

13. True or False: A digital audio sample cannot be converted back to the exact analog sound wave it was created from. Explain your answer. (0.5 points)

True; an audio sample cannot be converted back exactly because there is a loss of data from compression and as well by how much your sampling the audio a long with bit depth

14. Why does increasing the sampling rate improve the quality of an audio sample? (0.5 points)

Increasing the sampling rate improves the quality because then the sampling occurs more often taking more measurments from the signal being recieved

15. For the following questions, write out an RGB value as (x,y,z) with decimal numbers, assuming a bit depth of 8 per color. (1 point)

(a) What is white in RGB?

(255, 255, 255)

(b) What is black in RGB?

(0, 0, 0)

(c) List two RGB values that are similar to each other, but not exactly the same.

(255, 255, 240); (255, 255, 230)

(d) List two RGB values that are very different from each other.

(255, 0, 255); (255, 255, 0)